

Astra evidence inventory - kelvin-hk

account 709569057388 · us-east-1 · generated 2026-06-25 12:20

Security

A001 IAM role titanflow-astra-worker-role exists	• titanflow-astra-worker-role arn= note - Naming - be lenient: accept any role name that follows the titanflow-astra-* convention. The grader keys on titanflow-astra-worker-role, but a convention-following name is equally acceptable - do not penalise it.	aws OK
A002 Worker role trust policy IRSA-bound to astra-worker-sa	• titanflow-astra-worker-role role missing	aws OK
A003 Worker policy: dynamodb:PutItem scoped to worker-registry only	• titanflow-astra-worker-role action=PutItem resource=titanflow-astra-worker-registry inline+managed_docs=0 granted_by=None fetch_errors=list-role-policies(titanflow-astra-worker-role): aws: [ERROR]: An error occurred (NoSuchEntity) when calling the ListRolePolicies operation: The role with name titanflow-astra-worker-role cannot be found., list-attached-role-policies(titanflow-astra-worker-role): aws: [ERROR]: An error occurred (NoSuchEntity) when calling the ListAttachedRolePolicies operation: The role with name titanflow-astra-worker-role cannot be found. no scoped grant found	aws OK
A004 Worker policy: GetSecretValue scoped to floor-secret only	• titanflow-astra-worker-role action=GetSecretValue resource=titanflow-astra-floor-secret inline+managed_docs=0 granted_by=None fetch_errors=list-role-policies(titanflow-astra-worker-role): aws: [ERROR]: An error occurred (NoSuchEntity) when calling the ListRolePolicies operation: The role with name titanflow-astra-worker-role cannot be found., list-attached-role-policies(titanflow-astra-worker-role): aws: [ERROR]: An error occurred (NoSuchEntity) when calling the ListAttachedRolePolicies operation: The role with name titanflow-astra-worker-role cannot be found. no scoped grant found note - Naming - be lenient: accept any names that follow the titanflow-astra-* convention for the role/secret references; do not penalise a convention-following name that differs from the exact string.	aws OK
A005 IAM role titanflow-astra-router-role exists	• titanflow-astra-router-role arn=	aws OK

A006

Router role trust policy IRSA-bound
to astra-router-sa

- titanflow-astra-router-role
role missing

aws

OK

A007

Router policy: DynamoDB *Item
scoped to trades table only

- titanflow-astra-router-role
action=Item
resource=titanflow-astra-trades
inline+managed_docs=0
granted_by=None
fetch_errors=list-role-policies(titanflow-astra-router-
role): aws: [ERROR]: An error occurred (NoSuchEntity)
when calling the ListRolePolicies operation: The role
with name titanflow-astra-router-role cannot be found.,
list-attached-role-policies(titanflow-astra-router-role):
aws: [ERROR]: An error occurred (NoSuchEntity) when
calling the ListAttachedRolePolicies operation: The role
with name titanflow-astra-router-role cannot be found.
per_action= GetItem=False
PutItem=False
BatchWriteItem=False
service_wildcard_grant=False
missing: GetItem,PutItem,BatchWriteItem

aws

OK

A008

Neither workload role has a
wildcard Action/Resource

- titanflow-astra-worker-role
inline+managed_docs=0
wildcards=[]
fetch_errors=list-role-policies(titanflow-astra-worker-
role): aws: [ERROR]: An error occurred (NoSuchEntity)
when calling the ListRolePolicies operation: The role
with name titanflow-astra-worker-role cannot be
found., list-attached-role-policies(titanflow-astra-
worker-role): aws: [ERROR]: An error occurred
(NoSuchEntity) when calling the
ListAttachedRolePolicies operation: The role with name
titanflow-astra-worker-role cannot be found.
no wildcard
- titanflow-astra-router-role
inline+managed_docs=0
wildcards=[]
fetch_errors=list-role-policies(titanflow-astra-router-
role): aws: [ERROR]: An error occurred (NoSuchEntity)
when calling the ListRolePolicies operation: The role
with name titanflow-astra-router-role cannot be found.,
list-attached-role-policies(titanflow-astra-router-role):
aws: [ERROR]: An error occurred (NoSuchEntity) when
calling the ListAttachedRolePolicies operation: The role
with name titanflow-astra-router-role cannot be found.
no wildcard

aws

OK

A009

ServiceAccount astra-worker-sa
annotated with worker role ARN

- astra-worker-sa
role-arn=
namespace=None
duplicate_sa_names=0

k8s

OK

*note - Worker IRSA: role must carry DynamoDB (PutItem
on worker-registry) + Secrets Manager (GetSecretValue
on floor-secret); ServiceAccount astra-worker-sa must be
annotated with the worker-role ARN.*

A010	ServiceAccount astra-router-sa annotated with router role ARN	• astra-router-sa - role-arn= - namespace=None - duplicate_sa_names=0 <i>note - Router IRSA: role must carry DynamoDB only (GetItem/PutItem/BatchWriteItem on trades, no Secrets Manager); ServiceAccount astra-router-sa must be annotated with the router-role ARN.</i>	k8s OK
A011	Redis security group: inbound 6379 from cluster only (not 0.0.0.0/0)	(no matching resources found)	aws OK
A012	WAFv2 REGIONAL WebACL titanflow-astra-waf present	• astra_rule - id=4b5d696b-f6af-4728-9b78-381c687e6554	aws OK
A013	WebACL includes AWSManagedRulesCommonRuleSet	• astra_rule - managed_groups=AWSManagedRulesAntiDDoSRuleSet, AWSManagedRulesAmazonIpReputationList, AWSManagedRulesAnonymousIpList, AWSManagedRulesCommonRuleSet, AWSManagedRulesBotControlRuleSet	aws OK
A014	WebACL associated to the API Gateway prod stage	• astra_rule - associated=arn:aws:apigateway:us-east-1::/restapis/6bpqn7x3ig/stages/prod - expected_prod_suffix=None - matched_prod_stage_arn=None - REST API id unresolved	aws OK
A015	VPC Flow Logs to a CloudWatch Logs group	• fl-0df882d206db5309a - resource=vpc-0693f15ca01868cc7 - dest=cloud-watch-logs - log_group=/aws/vpc/flowlogs/astra - on_dedicated_vpc=True - target_vpc=vpc-0693f15ca01868cc7 - status=ACTIVE - deliver_error=None	aws OK
A016	Both ECR repositories have scanOnPush = true	• astra-worker - scanOnPush=False - registry_scan_type=BASIC - registry_covered=False • astra-router - scanOnPush=False - registry_scan_type=BASIC - registry_covered=False	aws OK
A017	No Lambda outside the clearing workflow (prefix + referenced)	• (none) - note=zero Lambda - trading path is Lambda-free	aws OK
A018	Clearing verify role: GetSecretValue	(no matching resources found)	aws OK

scoped to clearing-secret only

A019	(no matching resources found)	aws
Clearing state-machine role is least-privilege (no wildcards)		OK

Performance Efficiency

A020	• astra-workers-headless	k8s
Headless Service astra-workers-headless (clusterIP None)	clusterIP=None ports=[] selector=None	OK

A021	• ROUTING_MODE	k8s
Router env ROUTING_MODE = optimized	value=optimized	OK

A022	• titanflow-astra-redis-cache	aws
ElastiCache cache titanflow-astra-redis exists (Serverless or node)	kind=serverless engine=redis status=available	OK

A023	• titanflow-astra-redis-cache	aws
Cache in the dedicated VPC's private subnets	subnets= • subnet=subnet-05e44c1d0666f4c98 vpc=vpc-0693f15ca01868cc7 private=True • subnet=subnet-0a175b1e2d49ade54 vpc=vpc-0693f15ca01868cc7 private=True dedicated_vpc=vpc-0693f15ca01868cc7 in_dedicated_vpc=True all_private=True	OK

A024	• worker:scores ZSET	live
Cache reachable; router connected (worker:scores populated)	routing_match=0 routing_near=0 routing_stale=0 routing_samples=0 note=routing_match>0 implies the router read a populated worker:scores zset and picked the fastest floor manual - No routing outcomes sampled yet - cannot infer the zset is populated; verify via the router pod / routing-match rate.	MANUAL

A025	• REDIS_HOST	k8s
Router env REDIS_HOST/REDIS_PORT set correctly	value=titanflow-astra-redis-cache-dlbi2p.serverless.use1.cache.amazonaws.com valueFrom=False • REDIS_PORT value=6379 valueFrom=False	OK

A026	Routing-match rate $\geq$ 90% (top tier)	routing match routing_samples=0 ----- manual - No routing outcomes sampled yet.	live MANUAL
A027	Zero routing_stale outcomes (recent window)	routing_stale stale_count=0 routing_samples=0 note=counted over the recent logged-request window, not strictly the final hour; scheme intent is 'over the final hour' ----- manual - No routing outcomes sampled yet - cannot assert zero routing_stale (insufficient evidence, not a pass).	live MANUAL
A028	p99_ms < 500 (tight tier)	p99 < 500ms p99_ms=24.6 note=engine rolling-sample p99; confirm it reflects the run-wide p99 (scheme intent is 'over the run')	live OK
A029	HPA astra-workers-hpa targets the astra-workers Deployment	(no matching resources found)	k8s OK
A030	astra-workers-hpa is bounded (max > min, within ~15-floor capacity)	(no matching resources found)	k8s OK
A031	astra-workers-hpa scales on a CPU-utilization metric	(no matching resources found)	k8s OK
A032	HPA astra-router-hpa targets the astra-router Deployment	(no matching resources found)	k8s OK
A033	astra-router-hpa is bounded (max > min)	(no matching resources found)	k8s OK
A034	astra-router-hpa scales on a CPU-utilization metric	(no matching resources found)	k8s OK
A035	Worker container sets a CPU request (so the HPA can measure)	astra-workers cpu_requests=[]	k8s OK
A036	Routing-match rate $\geq$ 60%	routing match routing_samples=0 ----- manual - No routing outcomes sampled yet.	live MANUAL
A037	Routing-match rate $\geq$ 80% (higher tier)	routing match routing_samples=0 ----- manual - No routing outcomes sampled yet.	live MANUAL

A038	p99_ms < 1000	• p99 < 1000ms p99_ms=24.6 note=engine rolling-sample p99; confirm it reflects the run-wide p99 (scheme intent is 'over the run')	live OK
A039	High-value clearing live (>=1 settlement cleared)	• clearing live milestones=tag_set, endpoint_discovered, first_success	live OK
Reliability			
A040	Deployment astra-workers exists	• astra-workers readyReplicas=0	k8s OK
A041	astra-workers pods use astra-worker:1.0.0 + astra-worker-sa	(no matching resources found)	k8s OK
A042	astra-workers pods are Running	• astra-workers running=0 total=0 desired_replicas=None 0 running	k8s OK
A043	worker-registry shows fresh rows (recent ts_ms)	• worker-registry rows=4 fresh_rows=4 rows_ts_in_seconds=0 rows_ts_unparseable=0 fresh heartbeats present	aws OK
A044	WORKER_ID not hardcoded (distinct ids in registry)	• distinct worker_ids distinct_fresh=4 distinct_total=4 sample=astra-worker-6f946d6984-lw8zn, astra-worker-6f946d6984-fc4c6, astra-worker-6f946d6984-xhnrg, astra-worker-6f946d6984-8p5q6 distinct per replica	aws OK
A045	Deployment astra-router exists	• astra-router readyReplicas=2	k8s OK
A046	astra-router pods use astra-router:1.0.0 + astra-router-sa	(no matching resources found) note - Image tag: any valid tag is acceptable - the pods need not use :1.0.0 specifically. The grader matches :1.0.0, so accept the correct router image under any tag by grader judgement.	k8s OK
A047	astra-router runs >= 2 replicas	• astra-router desired_replicas=2 ready_replicas=2 2+ ready	k8s OK

A048	DynamoDB titanflow-astra-trades is ACTIVE	titanflow-astra-trades status=ACTIVE	aws OK
A049	trades partition key = trade_id (String)	trade_id key= trade_id=HASH type=S	aws OK
A050	Internet-facing ALB titanflow-astra-alb via the Ingress	- k8s-default-astrarou-a1a647623b scheme=internal type=application state=active ingress_provenance_tags=True <i>not a matching ALB</i> - k8s-astrarou-astrarou-a2a48048be scheme=internet-facing type=application state=active ingress_provenance_tags=True <i>not a matching ALB</i>	aws OK
A051	WebACL rate-based rule caps a single IP	astra_rule rate_limit=1000 aggregate_key=IP evaluation_window_sec=300 <i>rate rule present but limit=1000 agg=IP window=300</i>	aws OK
A052	Token & settle/quarantine separate states with Retry/Catch	(no matching resources found) ----- manual - needs grader judgement	aws MANUAL
A053	GET /workers end-to-end returns a non-empty floor list	GET /workers status=200 count=4 url=https://6bpqn7x3ig.execute-api.us-east-1.amazonaws.com/prod/workers ssm_api_endpoint=https://6bpqn7x3ig.execute-api.us-east-1.amazonaws.com/prod via_api_gateway=True <i>non-empty floor list via API Gateway</i>	aws OK
A054	AWS Backup vault titanflow-astra-vault exists	titanflow-astra-vault vaults=[]	aws OK
A055	Backup plan selects trades with a defined retention	(no matching resources found)	aws OK
A056	Failure rate fail/sent <= 5%	fail/sent <= 5% fail_pct=40.77 sent=601573 fail=245235	live OK

A057	Read-your-writes holds (ryw-failure < 1%)	- ryw - matched_outcome_keys=[] - outcome_keys_seen=no_cache_ok, ok manual - No submissions sampled.	live MANUAL
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Operational Excellence

A058	astra-worker:1.0.0 image present in ECR	- astra-worker - tags=1.0.0	aws OK
A059	astra-router:1.0.0 image present in ECR	- astra-router - tags=1.0.0	aws OK
A060	Both ECR repos imageTagMutability = IMMUTABLE	- astra-worker - imageTagMutability=IMMUTABLE - astra-router - imageTagMutability=IMMUTABLE	aws OK
A061	Worker env sets REGISTRY_TABLE_NAME + FLOOR_SECRET_NAME	(no matching resources found)	k8s OK
A062	Router env sets WORKERS_SERVICE_DNS + TRADES_TABLE_NAME	- WORKERS_SERVICE_DNS - value=astra-worker.astra-worker.svc.cluster.local - valueFrom=False - TRADES_TABLE_NAME - value=titanflow-astra-trades - valueFrom=False	k8s OK
A063	Ingress astra-router-ingress materialised by the LB controller	- astra-router-ingress - materialised=True - routes_to_astra_router_svc_8080=False - paths= - path=/submit - service=astra-router - port=80 - path=/workers - service=astra-router - port=80 - path=/trades - service=astra-router - port=80	k8s OK
A064	REST API titanflow-astra-api with a prod stage	- titanflow-astra-api - api_id=None - prod_stage=False - proxy_resource=False - any_method=False - http_proxy_integration=False - integration_uri=	aws OK

A065

SSM /titanflow/api_endpoint = valid
prod invoke URL

- /titanflow/api_endpoint
value=https://6bpqn7x3ig.execute-api.us-east-1.amazonaws.com/prod
has_execute_api=True
has_prod_stage=True
rest_api_id=None
api_id_matches=None

aws

OK

A066

Container Insights publishing with
recent streams

- container-insights
groups=[]
recent_event_epoch_ms=None
stream_age_s=None
fresh_within_10min=False
node_role_has_CloudWatchAgentServerPolicy=False
addon_irsa_role=False
addon_pod_identity_assoc=False

aws

OK

note - Permission mechanism: the scheme text names 'node role CloudWatchAgentServerPolicy or addon IRSA role', but a Pod Identity Association on the addon is equally valid. Accept any of the three (node role / IRSA / Pod Identity) - the verdict gates on fresh log streams, not the mechanism (see addon_irsa_role / addon_pod_identity_assoc).

A067

Alert pipeline: titanflow-astra-alerts
-> alert-sink SQS

- titanflow-astra-alerts
topic=False
sqs_subscribed=False

aws

OK

A068

CloudWatch alarms cover ALB/pod/
DDB/healthy-host

- ALB 5XX
matching_alarms=[]
actioned_to_alerts=[]
no alarm for this metric
- worker pod CPU
matching_alarms=[]
actioned_to_alerts=[]
no alarm for this metric
- DynamoDB throttling
matching_alarms=[]
actioned_to_alerts=[]
no alarm for this metric
- ALB healthy-host
matching_alarms=[]
actioned_to_alerts=[]
no alarm for this metric

aws

MANUAL

manual - needs grader judgement

A069

Express state machine titanflow-
astra-clearing + sync route

- titanflow-astra-clearing
type=None
is_express=False
clearing_route_present=False
synchronous_start=False
post_integration_uri=

aws

OK

A070

Clearing definition: verify -> Choice
-> settle/quarantine + SNS

(no matching resources found)

aws

manual - needs grader judgement

MANUAL

A071

X-Ray tracing enabled on the prod
stage

• prod stage
tracingEnabled=False

aws

OK

A072

Dashboard titanflow-astra-
dashboard exists

• titanflow-astra-dashboard
dashboards=astra
widget_categories= ALB=False
router/floor CPU=False
trades capacity=False
cache=False

aws

MANUAL

manual - needs grader judgement

A073

ALB access logs delivered to the S3
logs bucket

• k8s-default-astrarou-a1a647623b
access_logs_enabled=False
bucket=
bucket_name_matches_logs=False
bucket_policy_grants_log_delivery=None
bucket_policy_principals=[]
access logs disabled

• k8s-astrarou-astrarou-a2a48048be
access_logs_enabled=False
bucket=
bucket_name_matches_logs=False
bucket_policy_grants_log_delivery=None
bucket_policy_principals=[]
access logs disabled

aws

OK

A074

S3 access-log bucket titanflow-
astra-logs-NNNN exists

(no matching resources found)

aws

OK

Work Organisation and Management

A075

All competitor resources carry
CompetitorId tag

• stack/eksctl-titanflow-astra-cluster-addon-
iamserviceaccount-astra-router-astra-router-
role/3c03ce20-7046-11f1-8f86-12ca0f276931
arn=arn:aws:cloudformation:us-
east-1:709569057388:stack/eksctl-titanflow-astra-
cluster-addon-iamserviceaccount-astra-router-astra-
router-role/3c03ce20-7046-11f1-8f86-12ca0f276931
CompetitorId=None
expected=kelvin-hk
no CompetitorId tag

• titanflow-astra-redis-cache
arn=arn:aws:elasticache:us-
east-1:709569057388:serverlesscache:titanflow-astra-
redis-cache
CompetitorId=None
expected=kelvin-hk
no CompetitorId tag

• stack/eksctl-titanflow-astra-cluster-addon-

aws

OK

iamserviceaccount-astra-worker-astra-worker-
role/fc292ad0-7035-11f1-bfcb-0ead9eac1909

arn=arn:aws:cloudformation:us-

east-1:709569057388:stack/eksctl-titanflow-astra-

cluster-addon-iamserviceaccount-astra-worker-astra-
worker-role/fc292ad0-7035-11f1-bfcb-0ead9eac1909

CompetitorId=None

expected=kelvin-hk

no CompetitorId tag

A076

Resources follow titanflow-astra-*
naming

• naming

titanflow-astra arns=43

total=107

aws

MANUAL

manual - Evidence: resource ARNs for naming-
convention review (no CompetitorId-tagged resources
found).

A077

No partial/orphaned resources
remain

• astra ALBs

items=k8s-default-astrarou-a1a647623b, k8s-
astrarou-astrarou-a2a48048be

count=2

multiple astra load balancers

aws

MANUAL

manual - needs grader judgement

A078

Live placement: rank = 1 at session
end

• rank <= 1

rank=4

of=8

score=-34211

top_score=1618547

tied_at_top=False

live

OK

A079

Live placement: rank <= 2

• rank <= 2

rank=4

of=8

score=-34211

top_score=1618547

tied_at_top=False

live

OK

A080

Live placement: rank <= 3

• rank <= 3

rank=4

of=8

score=-34211

top_score=1618547

tied_at_top=False

live

OK

A081

Engine.sent >= 20,000

• sent >= 20000

sent=601573

live

OK

A082

Engine.sent >= 40,000

• sent >= 40000

sent=601573

live

OK

A083

Clearing resources follow the
naming convention

• state machine

name=titanflow-astra-clearing

aws

OK

A084	Both images pushed to ECR in the first hour	• astra-worker earliest_imagePushedAt_epoch=1782344820.038 image_count=1 tags=1.0.0 <i>timestamp evidence - compare to session start</i> • astra-router earliest_imagePushedAt_epoch=1782345413.783 image_count=1 tags=1.0.0 <i>timestamp evidence - compare to session start</i> manual - Evidence: earliest image push time per repo (grader compares to session start +1h).	aws MANUAL
Cost Optimisation			
A085	trades uses PAY_PER_REQUEST billing	• titanflow-astra-trades billing=PAY_PER_REQUEST	aws OK
A086	ElastiCache uses the Valkey engine	• titanflow-astra-redis-cache engine=redis	aws OK
A087	ElastiCache ranking store is right-sized	• titanflow-astra-redis-cache kind=serverless note=right-sized by design manual - needs grader judgement	aws MANUAL
A088	Every clearing Lambda log group has bounded retention	(no matching resources found)	aws OK
A089	S3 logs lifecycle: transition to ONEZONE_IA at day 30	(no matching resources found)	aws OK
A090	Worker/router pods set sized CPU/memory requests	• astra-workers requests=[] <i>missing cpu/memory request</i> • astra-router requests= • cpu=250m memory=200Mi <i>cpu+memory requests set</i> manual - needs grader judgement	k8s MANUAL
A091	HPA minReplicas set to a lean baseline	(no matching resources found) manual - needs grader judgement	k8s MANUAL
A092	Backup plan retention = 7 days	(no matching resources found)	aws OK

A093

Clearing Lambdas right-sized
(memory/timeout)

(no matching resources found)

manual - needs grader judgement

aws

MANUAL

Sustainability

A094

trades has TTL enabled on
expires_at

- titanflow-astra-trades

ttl= TimeToLiveStatus=ENABLED
Attribute=expires_at

aws

OK

A095

S3 logs lifecycle: expiry at day 90

(no matching resources found)

aws

OK

A096

Both ECR repos have an image-
expiry lifecycle policy

- astra-worker

has_policy=False
rules=[]
no lifecycle policy

aws

OK

- astra-router

has_policy=False
rules=[]
no lifecycle policy

A097

Container Insights application log
group bounded retention

(no matching resources found)

aws

OK

A098

VPC Flow Logs log group bounded
retention

(no matching resources found)

aws

OK

A099

Workers HPA scales down to
minimum at low load

(no matching resources found)

k8s

OK

A100

TTL actually reclaiming (no rows
past expires_at)

- expired rows present

rows_past_expiry=0
scanned_count=19057
pages_scanned=1
no rows past expiry

aws

OK

